



HMCAD1520

v04.1015



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

Features

- High Speed Modes (12-bit / 8-bit)
 - Quad Channel Mode: $F_{Smax} = 160 / 250$ MSPS
 - Dual Channel Mode: $F_{Smax} = 320 / 500$ MSPS
 - Single Channel Mode: $F_{Smax} = 640 / 1000$ MSPS
 - SNR: 70 dB, SFDR: 60/75 dB [1] (12-bit 1ch Mode)
- 8-bit Modes Described in HMCAD1511 and HMCAD1510
- Precision Mode (14-bit)
 - Four channels up to 105 MSPS
 - SNR: 74 dB, SFDR: 83 dB @ 70 MHz
 - SNR: 72.5 dB, SFDR: 78 dB @ 140 MHz
- Integrated Cross Point Switches with instantaneous switching
- Internal low jitter programmable Clock Divider
- Ultra Low Power Dissipation
 - 490 mW including I/O at 12-bit 640 MSPS
- 0.5 μ s start-up time from Sleep, 15 μ s from Power Down
- Internal reference circuitry with no external components required
- Coarse and fine gain control

- Digital fine gain adjustment for each ADC
- Internal offset correction
- 1.8 V supply voltage
- 1.7 - 3.6 V CMOS logic on control interface pins
- Serial LVDS output
 - 12, 14, 16 and Dual 8-bit modes available
- 7 x 7 mm 48 QFN Package

[1] Including/Excluding Interleaving Spurs

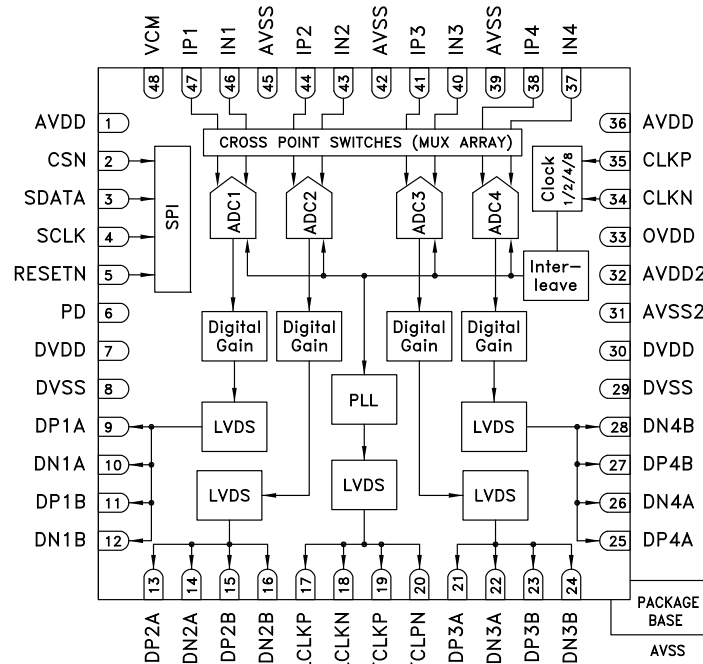
Typical Applications

- Precision Oscilloscopes
- Spectrum Analyzers
- Diversity Receivers
- Hi-End Ultrasound
- Communication Testing
- Non Destructive Testing

Pin compatible parts

HMCAD1520 is pin compatible and can be configured to operate as HMCAD1511 and HMCAD1510, with functionality and performance as described in HMCAD1511 and HMCAD1510 datasheets.

Functional Diagram



Functional Block Diagram

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HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

General Description

The HMCAD1520 is a versatile high performance low power analog-to-digital converter (ADC), with interleaving High Speed Modes to increase sampling rate. Integrated Cross Point Switches activate the input selected by the user.

In Single Channel Mode, one of the four inputs can be selected as valid input to the single ADC channel. In Dual Channel Mode, any two of the four inputs can be selected to each ADC channel. In Quad Channel Mode and Precision Mode, any input can be assigned to any ADC channel.

An internal, low jitter and programmable clock divider makes it possible to use a single clock source for all operational modes.

The HMCAD1520 is based on a proprietary structure, and employs internal reference circuitry, a serial control interface and a serial LVDS output data. Data and frame synchronization clocks are supplied for data capture at the receiver. Internal digital fine gain can be set separately for each ADC to calibrate for gain errors.

Various modes and configuration settings can be applied to the ADC through the serial control interface (SPI). Each channel can be powered down independently and output data format can be selected through this interface. A full chip idle mode can be set by a single external pin. Register settings determine the exact function of this pin.

HMCAD1520 is designed to interface easily with Field Programmable Gate Arrays (FPGAs) from several vendors.

Electrical Specifications

DC Specifications

AVDD = DVDD = OVDD = 1.8V, $F_s = 160$ MSPS, Quad Channel 12-bit High Speed Mode, 50% Clock Duty Cycle, -1 dBFS 70 MHz Input Signal, 1x / 0 dB Digital Gain (Fine and Coarse), Unless Otherwise Noted

Parameter	Description	Min	Typ	Max	Unit
DC accuracy					
No missing codes		Guaranteed			
Offset	Offset error after internal digital offset correction		1		LSB
G_{abs}	Gain error			±6	%FS
G_{rel}	Gain matching between channels. ±3 sigma value at worst case conditions		±0.5		%FS
DNL	Differential non linearity		±0.2		LSB
INL	Integral non linearity		±0.6		LSB
$V_{CM,out}$	Common mode voltage output		$V_{AVDD}/2$		
Analog Input					
$V_{CM,in}$	Analog input common mode voltage	$V_{CM}-0.1$		$V_{CM}+0.2$	V
FSR	Differential input voltage full scale range		2		Vpp
$C_{in,Q}$	Differential input capacitance, Quad channel mode		5		pF
$C_{in,D}$	Differential input capacitance, Dual channel mode		7		pF
$C_{in,S}$	Differential input capacitance, Single channel mode		11		pF
Power Supply					
V_{AVDD}	Analog Supply Voltage	1.7	1.8	2	V
V_{DVDD}	Digital and output driver supply voltage	1.7	1.8	2	V
V_{OVDD}	Digital CMOS Input Supply Voltage	1.7	1.8	3.6	V
Temperature					
T_A	Operating free-air temperature	-40		85	°C

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A / D CONVERTERS - SMT



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

AC Specifications – High Speed Modes

AVDD = DVDD = OVDD = 1.8V, 50% clock duty cycle, -1 dBFS 70 MHz input signal, Gain = 1X, 12-bit output, RSDS output data levels, unless otherwise noted

Parameter	Description	Min	Typ	Max	Unit
Performance					
SNR	Signal to Noise Ratio, excluding interleaving spurs				
	Single Channel Mode, $F_s = 640$ MSPS		70		dBFS
	Dual Channel Mode, $F_s = 320$ MSPS		70		dBFS
	Quad Channel Mode, $F_s = 160$ MSPS		70		dBFS
SINAD _{incl}	Signal to Noise and Distortion Ratio, including interleaving spurs				
	Single Channel Mode, $F_s = 640$ MSPS		58		dBFS
	Dual Channel Mode, $F_s = 320$ MSPS		58		dBFS
	Quad Channel Mode, $F_s = 160$ MSPS		58		dBFS
SINAD _{excl}	Signal to Noise and Distortion Ratio, excluding interleaving spurs				
	Single Channel Mode, $F_s = 640$ MSPS		67		dBFS
	Dual Channel Mode, $F_s = 320$ MSPS		68		dBFS
	Quad Channel Mode, $F_s = 160$ MSPS		68		dBFS
SFDR _{incl}	Spurious Free Dynamic Range, including interleaving spurs				
	Single Channel Mode, $F_s = 640$ MSPS		60		dBc
	Dual Channel Mode, $F_s = 320$ MSPS		60		dBc
	Quad Channel Mode, $F_s = 160$ MSPS		60		dBc
SFDR _{excl}	Spurious Free Dynamic Range, excluding interleaving spurs				
	Single Channel Mode, $F_s = 640$ MSPS		75		dBc
	Dual Channel Mode, $F_s = 320$ MSPS		77		dBc
	Quad Channel Mode, $F_s = 160$ MSPS		78		dBc
HD2/3	Worst of HD2/HD3				
	Single Channel Mode, $F_s = 640$ MSPS		75		dBc
	Dual Channel Mode, $F_s = 320$ MSPS		77		dBc
	Quad Channel Mode, $F_s = 160$ MSPS		78		dBc
ENOB	Effective number of Bits				
	Single Channel Mode, $F_s = 640$ MSPS		10.8		bits
	Dual Channel Mode, $F_s = 320$ MSPS		11.0		bits
	Quad Channel Mode, $F_s = 160$ MSPS		11.0		bits
$X_{\text{tik,HS2}}$	CrossTalk Dual Ch Mode. Signal applied to 1 channel (F_{IN0}). Measurement taken on one channel with full scale at F_{IN1} , $F_{\text{IN1}} = 71$ MHz, $F_{\text{IN0}} = 70$ MHz		70		dBc
$X_{\text{tik,HS4}}$	CrossTalk Quad Ch Mode. Signal applied to 1 channel (F_{IN0}). Measurement taken on one channel with full scale at F_{IN1} , $F_{\text{IN1}} = 71$ MHz, $F_{\text{IN0}} = 70$ MHz		70		dBc
Power Supply	Single Ch: $F_s = 640$ MSPS, Dual Ch: $F_s = 320$ MSPS, Quad Ch: $F_s = 160$ MSPS.				
I_{AVDD}	Analog Supply Current		190		mA
I_{DVDD}	Digital and output driver Supply Current		82		mA
P_{AVDD}	Analog Power		342		mW

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HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

AC Specifications – High Speed Modes

AVDD = DVDD = OVDD = 1.8V, 50% clock duty cycle, -1 dBFS 70 MHz input signal, Gain = 1X, 12-bit output, RSDS output data levels, unless otherwise noted

Parameter	Description	Min	Typ	Max	Unit
P _{DVDD}	Digital Power		148		mW
P _{TOT}	Total Power Dissipation		490		mW
P _{PD}	Power Down Mode Dissipation		15		μW
P _{SLP}	Deep Sleep Mode Power Dissipation		66		mW
P _{SLPCH}	Power Dissipation with all channels in sleep channel mode (Light Sleep)		121		mW
P _{SLPCH_SAV}	Power Dissipation savings per channel off		92		mW
Analog Input					
FPBW	Full Power Bandwidth		700		MHz
Clock Inputs					
F _{Smax}	Max. Conversion Rate in Modes: Single / Dual Quad Channel	640/320 160			MSPS
F _{Smin}	Min. Conversion Rate in Modes: Single / Dual Quad Channel			120/60 30	MSPS

AC Specifications – Precision Mode

AVDD = DVDD = OVDD = 1.8V, FS = 105 MHz, 50% clock duty cycle, -1 dBFS 70 MHz input signal, Gain = 1X, dual 8-bit output, RSDS output data levels, unless otherwise noted

Parameter	Description	Min	Typ	Max	Unit
Performance					
SNR	Signal to Noise Ratio				
	F _S = 80 MSPS		75		dBFS
	F _S = 105 MSPS		74		dBFS
	F _S = 105 MSPS, F _{in} = 105 MSPS		72.5		dBFS
SINAD	Signal to Noise and Distortion Ratio				
	F _S = 80 MSPS		73		dBFS
	F _S = 105 MSPS		72.5		dBFS
	F _S = 105 MSPS, F _{in} = 105 MSPS		71		dBFS
SFDR	Spurious Free Dynamic Range				
	F _S = 80 MSPS		85		dBc
	F _S = 105 MSPS		83		dBc
	F _S = 105 MSPS, F _{in} = 105 MSPS		78		dBc
HD2	Second order harmonic spur				
	F _S = 80 MSPS		90		dBc
	F _S = 105 MSPS		90		dBc
	F _S = 105 MSPS, F _{in} = 105 MSPS		80		dBc
HD3	Third order harmonic spur				



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

AC Specifications – Precision Mode

AVDD = DVDD = OVDD = 1.8V, FS = 105 MHz, 50% clock duty cycle, -1 dBFS 70 MHz input signal, Gain = 1X, dual 8-bit output, RSDS output data levels, unless otherwise noted

Parameter	Description	Min	Typ	Max	Unit
	$F_s = 80$ MSPS		85		dBc
	$F_s = 105$ MSPS		83		dBc
	$F_s = 105$ MSPS, $F_{in} = 105$ MSPS		78		dBc
ENOB	Effective number of Bits				
	$F_s = 80$ MSPS		11.8		bits
	$F_s = 105$ MSPS		11.8		bits
	$F_s = 105$ MSPS, $F_{in} = 105$ MSPS		11.5		bits
X_{tik}	Crosstalk. Signal applied to 1 channel (F_{in0}). Measurement taken on one channel with full scale at F_{in1} . $F_{in1} = 71$ MHz, $F_{in0} = 70$ MHz		70		dBc
Power Supply					
I_{AVDD}	Analog Supply Current		229		mA
I_{DVDD}	Digital and output driver Supply Current		106		mA
P_{AVDD}	Analog Power		412		mW
P_{DVDD}	Digital Power		191		mW
P_{TOT}	Total Power Dissipation		603		mW
P_{PD}	Power Down Mode Dissipation		15		μW
P_{SLP}	Deep Sleep Mode Power Dissipation		66		mW
P_{SLPCH}	Power Dissipation with all channels in sleep channel mode (Light Sleep)		131		mW
P_{SLPCH_SAV}	Power Dissipation savings per channel off		118		mW
Analog Input					
FPBW	Full Power Bandwidth		700		MHz
Clock Inputs					
F_{Smax}	Max. Conversion Rate	105			MSPS
F_{Smin}	Min. Conversion Rate			15	MSPS

Digital and Switching Specifications

AVDD = DVDD = OVDD = 1.8V, RSDS output data levels, unless otherwise noted.

Parameter	Description	Min	Typ	Max	Unit
Clock Inputs					
DC	Duty Cycle, High speed modes	40		60	% high
DC	Duty Cycle, Precision mode	30		70	% high
Compliance	LVDS supported up to 700 Mbps	LVPECL, Sine wave, CMOS, LVDS			
$V_{CK,sine}$	Differential input voltage swing, sine wave clock input	1500			mVpp
$V_{CK,CMOS}$	Voltage input range CMOS (CLKN connected to ground)		V_{OVDD}		
$V_{CM,CK}$	Input common mode voltage. Keep voltages within ground and voltage of OVDD	0.3		$V_{OVDD} - 0.3$	V
C_{CK}	Differential Input capacitance		3		pF
Logic inputs (CMOS)					

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Digital and Switching Specifications

AVDD = DVDD = OVDD = 1.8V, RSDS output data levels, unless otherwise noted.

Parameter	Description	Min	Typ	Max	Unit
V _{HI}	High Level Input Voltage. V _{OVDD} ≥ 3.0V	2			V
V _{HI}	High Level Input Voltage. V _{OVDD} = 1.7V – 3.0V	0.8 · V _{OVDD}			V
V _{LI}	Low Level Input Voltage. V _{OVDD} ≥ 3.0V	0		0.8	V
V _{LI}	Low Level Input Voltage. V _{OVDD} = 1.7V – 3.0V	0		0.2 · V _{OVDD}	V
I _{HI}	High Level Input leakage Current			+/-10	μA
I _{LI}	Low Level Input leakage Current			+/-10	μA
C _I	Input Capacitance		3		pF
Data outputs					
Compliance		LVDS / RSDS			
V _{OUT}	Differential output voltage, LVDS		350		mV
V _{OUT}	Differential output voltage, RSDS		150		mV
V _{CM}	Output common mode voltage		1.2		V
Output coding	Default/optional	Offset Binary/ 2's complement			
Timing Characteristics					
t _{A,HS}	Aperture delay, High speed modes		1.5		ns
t _{A,PM}	Aperture delay, Precision mode		1.4		ns
t _{j,HS}	Aperture jitter, all bits set to '1' in jitter_ctrl<7:0>, High speed modes		120		fsrms
t _{j,HS}	Aperture jitter, one bit set to '1' in jitter_ctrl<7:0>, High speed modes		160		fsrms
t _{j,PM}	Aperture jitter, all bits set to '1' in jitter_ctrl<7:0>, Precision modes		75		fsrms
t _{j,PM}	Aperture jitter, one bit set to '1' in jitter_ctrl<7:0>, Precision modes		130		fsrms
T _{skew}	Timing skew between ADC channels, High speed modes		2.5		psrms
T _{SU}	Start up time from Power Down Mode and Deep Sleep Mode to Active Mode in μs. See section "Clock Frequency" for details.		15		μs
T _{SLPCH}	Start up time from Sleep Channel Mode to Active Mode		0.5		μs
T _{OVR}	Out of range recovery time		1		clock cycles
T _{LATPM}	Pipeline delay, Precision Speed Mode	15			clock cycles
T _{LATHSMQ}	Pipeline delay, Quad High Speed Mode	32			clock cycles
T _{LATHSMD}	Pipeline delay, Dual High Speed Mode	64			clock cycles
T _{LATHSMS}	Pipeline delay, Single High Speed Mode	128			clock cycles
LVDS Output Timing Characteristics					
t _{data}	LCLK to data delay time (excluding programmable phase shift)		50		ps
T _{PROP}	Clock propagation delay.	6·T _{LVDS} +2.2	7·T _{LVDS} +3.5	7·T _{LVDS} +5.0	ns
	LVDS bit-clock duty-cycle	45		55	% LCLK cycle
	Frame clock cycle-to-cycle jitter			2.5	% LCLK cycle
T _{EDGE}	Data rise- and fall time 20% to 80%		0.7		ns
T _{CLKEDGE}	Clock rise- and fall time 20% to 80%		0.7		ns

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A / D CONVERTERS - SMT



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

Absolute Maximum Ratings

Applying voltages to the pins beyond those specified in Table 1 could cause permanent damage to the circuit.

Table 1: Maximum voltage ratings

Pin	Reference pin	Rating
AVDD	AVSS	-0.3V to +2.3V
DVDD	DVSS	-0.3V to +2.3V
OVDD	AVSS	-0.3V to +3.9V
AVSS / DVSS	DVSS / AVSS	-0.3V to +0.3V
Analog inputs and outputs	AVSS	-0.3V to +2.3V
CLKx	AVSS	-0.3V to +3.9V
LVDS outputs	DVSS	-0.3V to +2.3V
Digital inputs	DVSS	-0.3V to +3.9V

Table 2 shows the maximum external temperature ratings.

Table 2: Maximum Temperature Ratings

Operating Temperature	-40 to +85 °C
Storage Temperature	-60 to +150 °C
Maximum Junction Temperature	110 °C
Thermal Resistance (Rth)	29 °C/W
Soldering Profile Qualification	J-STD-020
ESD Sensivity HBM	Class 1C
ESD Sensivity CDM	Class III



**ELECTROSTATIC SENSITIVE DEVICE
OBSERVE HANDLING PRECAUTIONS**

Stresses above those listed under Absolute Maximum Ratings may cause permanent damage to the device. This is a stress rating only; functional operation of the device at these or any other conditions above those indicated in the operational section of this specification is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.



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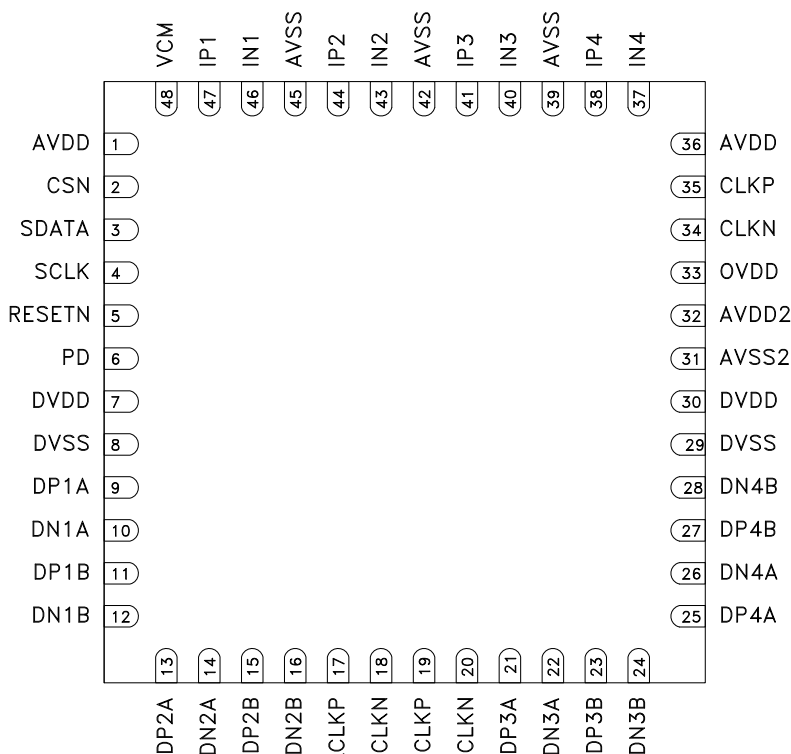


Figure 1: Pin Diagram

Table 3: Pin Descriptions

Pin name	Description	Pin Number	# of Pins
AVDD	Analog power supply, 1.8V	1, 36	2
CSN	Chip select enable. Active low	2	1
SDATA	Serial data input	3	1
SCLK	Serial clock input	4	1
RESETN	Reset SPI interface. Active low	5	1
PD	Power-down input. Activate after applying power in order to initialize the ADC correctly. Alternatively use the SPI power down feature	6	1
DVDD	Digital and I/O power supply, 1.8V	7, 30	2
DVSS	Digital ground	8, 29	2
DP1A	LVDS channel 1A, positive output	9	1
DN1A	LVDS channel 1A, negative output	10	1
DP1B	LVDS channel 1B, positive output	11	1
DN1B	LVDS channel 1B, negative output	12	1
DP2A	LVDS channel 2A, positive output	13	1
DN2A	LVDS channel 2A, negative output	14	1
DP2B	LVDS channel 2B, positive output	15	1
DN2B	LVDS channel 2B, negative output	16	1



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

Table 3: Pin Descriptions

Pin name	Description	Pin Number	# of Pins
LCLKP	LVDS bit clock, positive output	17	1
LCLKN	LVDS bit clock, negative output	18	1
FCLKP	LVDS frame clock (1X), positive output	19	1
FCLKN	LVDS frame clock (1X), negative output	20	1
DP3A	LVDS channel 3A, positive output	21	1
DN3A	LVDS channel 3A, negative output	22	1
DP3B	LVDS channel 3B, positive output	23	1
DN3B	LVDS channel 3B, negative output	24	1
DP4A	LVDS channel 4A, positive output	25	1
DN4A	LVDS channel 4A, negative output	26	1
DP4B	LVDS channel 4B, positive output	27	1
DN4B	LVDS channel 4B, negative output	28	1
AVSS2	Analog ground domain 2	31	1
AVDD2	Analog power supply domain 2, 1.8V	32	1
OVDD	Digital CMOS Inputs supply voltage	33	1
CLKN	Negative differential input clock.	34	1
CLKP	Positive differential input clock	35	1
IN4	Negative differential input signal, channel 4	37	1
IP4	Positive differential input signal, channel 4	38	1
AVSS	Analog ground	39, 42, 45	3
IN3	Negative differential input signal, channel 3	40	1
IP3	Positive differential input signal, channel 3	41	1
IN2	Negative differential input signal, channel 2	43	1
IP2	Positive differential input signal, channel 2	44	1
IN1	Negative differential input signal, channel 1	46	1
IP1	Positive differential input signal, channel 1	47	1
VCM	Common mode output pin, 0.5*AVDD	48	1

Start up Initialization

As part of the HMCAD1520 power-on sequence both a reset and a power down cycle have to be applied to ensure correct start-up initialization. Reset can be done in one of two ways:

1. By applying a low-going pulse (minimum 20 ns) on the RESETN pin (asynchronous).
2. By using the serial interface to set the 'rst' bit high. Internal registers are reset to default values when this bit is set. The 'rst' bit is self-reset to zero. When using this method, do not apply any low-going pulse on the RESETN pin.

Power down cycling can be done in one of two ways:

1. By applying a high-going pulse (minimum 20 ns) on the PD pin (asynchronous).
2. By cycling the 'pd' bit in register 0Fhex to high (reg value '0200'hex) and then low (reg value '0000'hex).



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

Register Initialization

To set the HMCAD1520 in Precision Mode, the following registers must be changed from the default value. Suggested values are:

Address	Data	Function
0x31	0x0008	Sets HMCAD1520 in precision mode, Clock divider to 1
0x53	0x0004	Sets the LVDS output in dual 8 bit mode

Serial Interface

The HMCAD1520 configuration registers can be accessed through a serial interface formed by the pins SDATA (serial interface data), SCLK (serial interface clock) and CSN (chip select, active low). The following occurs when CSN is set low:

- Serial data are shifted into the chip
- At every rising edge of SCLK, the value present at SDATA is latched
- SDATA is loaded into the register every 24th rising edge of SCLK

Multiples of 24-bit words data can be loaded within a single active CSN pulse. If more than 24 bits are loaded into SDATA during one active CSN pulse, only the first 24 bits are kept. The excess bits are ignored. Every 24-bit word is divided into two parts:

- The first eight bits form the register address
- The remaining 16 bits form the register data

Acceptable SCLK frequencies are from 20MHz down to a few hertz. Duty-cycle does not have to be tightly controlled.

Timing Diagram

Figure 2 shows the timing of the serial port interface. Table 4 explains the timing variables used in figure 2.

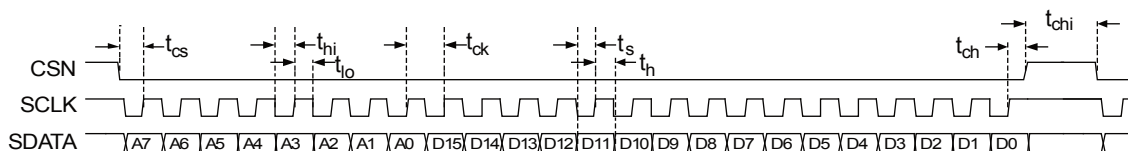


Figure 2: Serial Port Interface timing

Table 4: Serial Port Interface Timing Definitions

Parameter	Description	Minimum value	Unit
t_{cs}	Setup time between CSN and SCLK	8	ns
t_{ch}	Hold time between CSN and SCLK	8	ns
t_{hi}	SCLK high time	20	ns
t_{lo}	SCLK low time	20	ns
t_{ck}	SCLK period	50	ns
t_s	Data setup time	5	ns
t_h	Data hold time	5	ns



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

Timing Diagrams

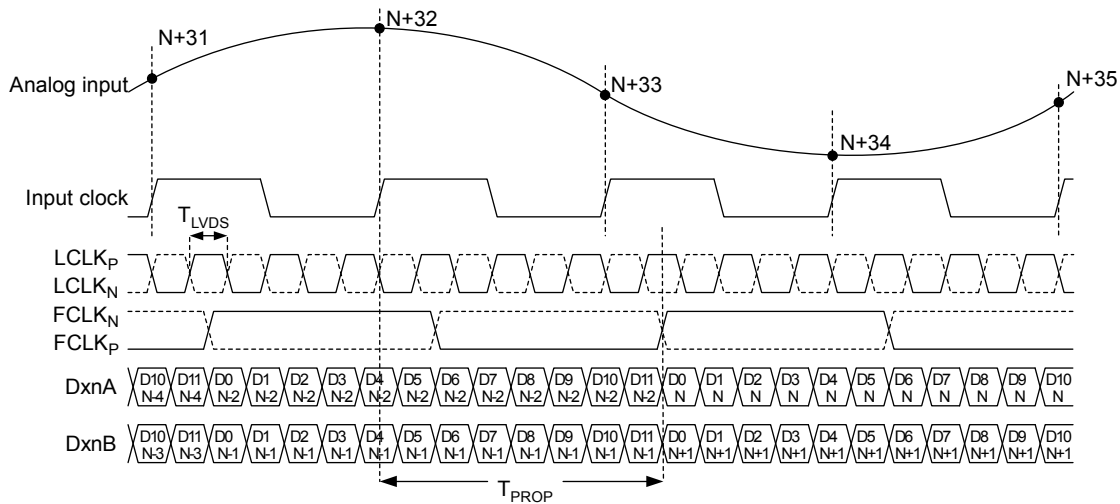


Figure 3: Quad channel - LVDS timing 12-bit output

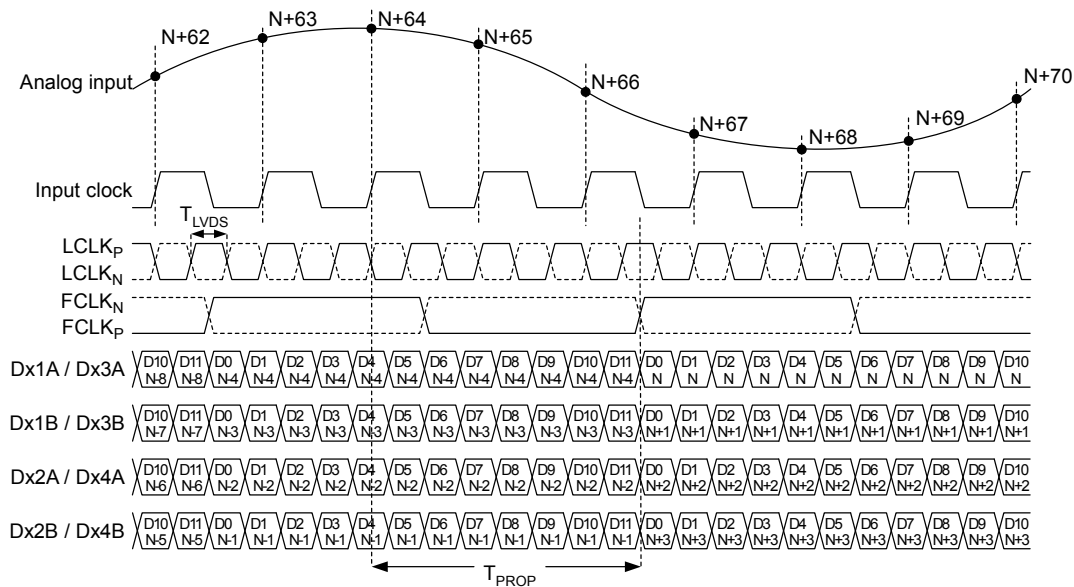


Figure 4: Dual channel - LVDS timing 12-bit output



**HIGH SPEED MULTI-MODE 8/12/14-BIT
1000/640/105 MSPS A/D CONVERTER**

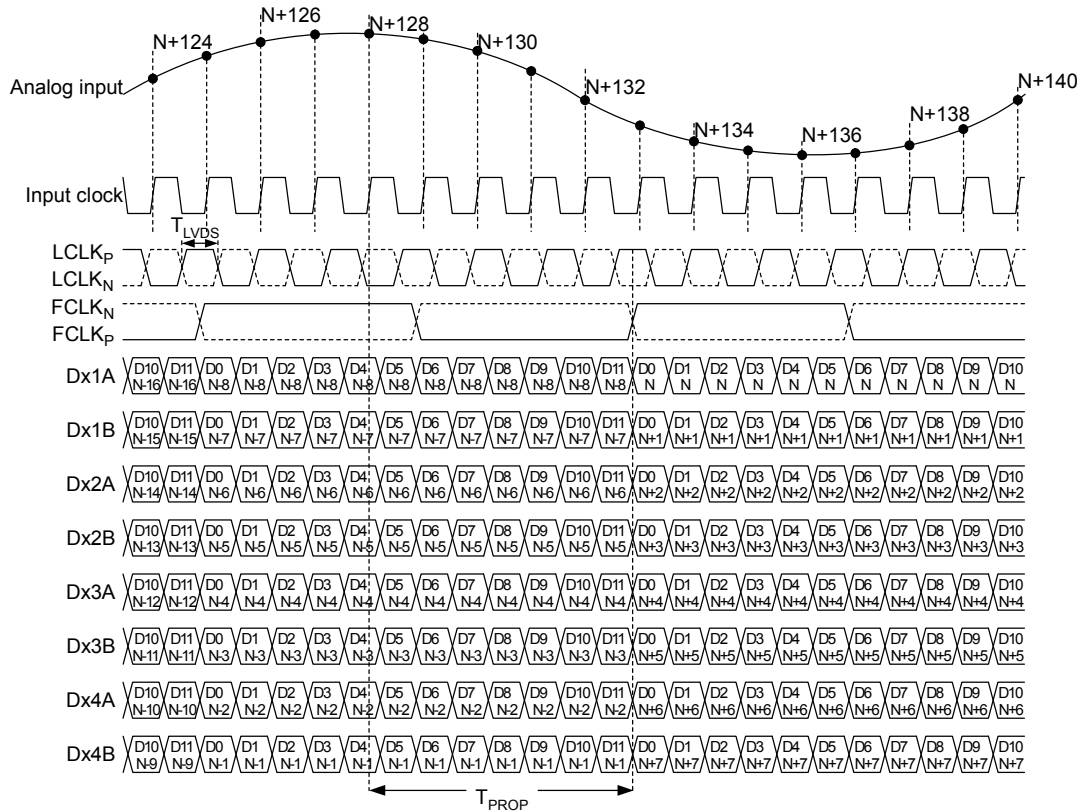


Figure 5: Single channel - LVDS timing 12-bit output

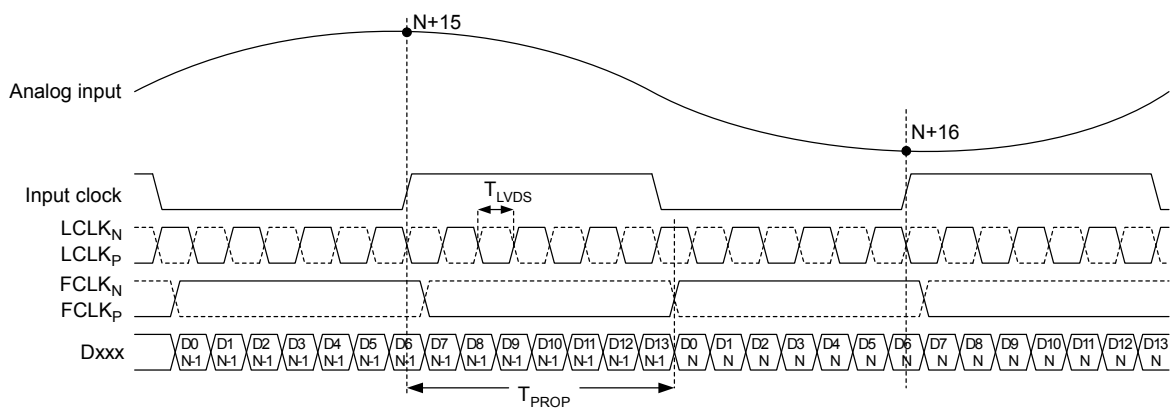


Figure 6: Precision - LVDS timing 14-bit output



**HIGH SPEED MULTI-MODE 8/12/14-BIT
1000/640/105 MSPS A/D CONVERTER**

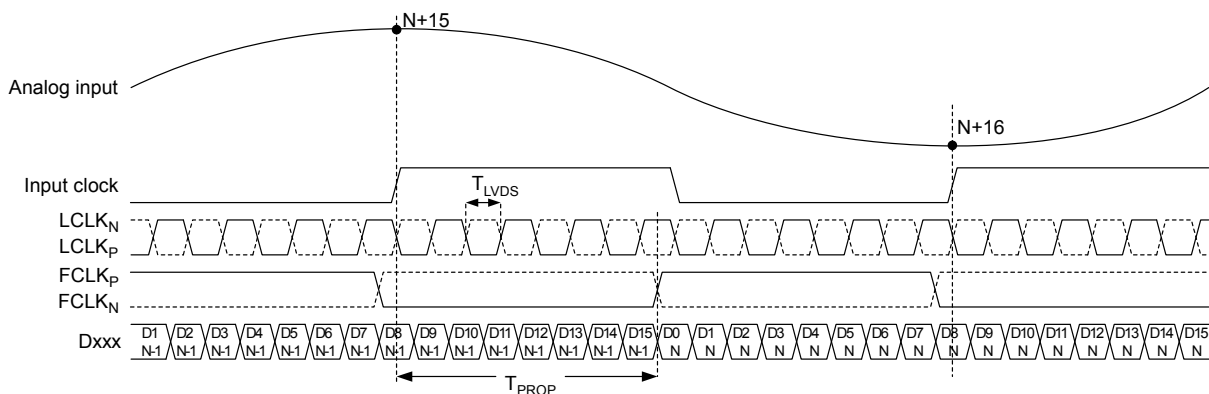


Figure 7: Precision - LVDS timing 16-bit output

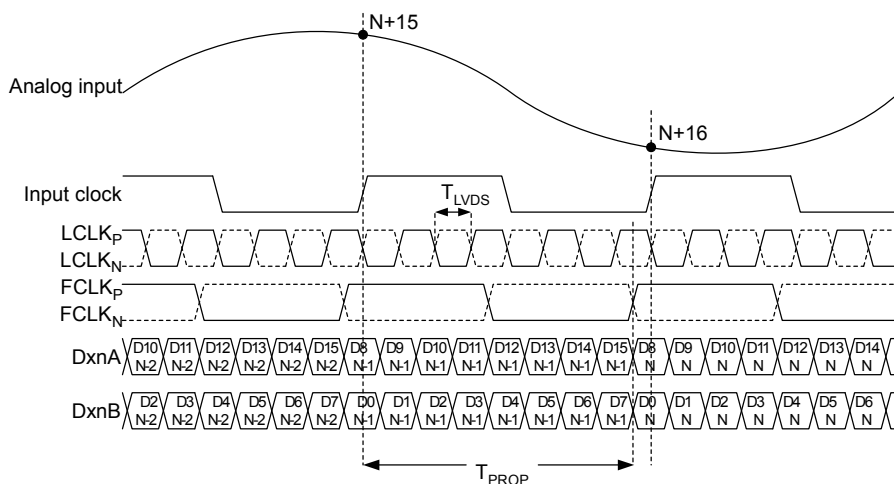


Figure 8: Precision - LVDS timing Dual 8-bit output

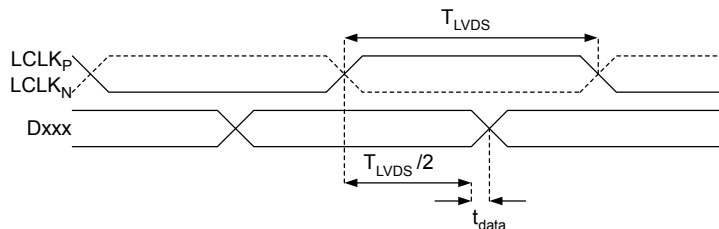


Figure 9: LVDS data timing



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

Table 5: Register Map

Name	Description	Default	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Hex Address
rst *	Self-clearing software reset.	Inactive																X	0x00
sleep4_ch <4:1>	Channel-specific sleep mode for a Quad Channel setup.	Inactive													X	X	X	X	0x0F
sleep2_ch <2:1>	Channel-specific sleep mode for a Dual Channel setup.	Inactive											X	X					
sleep1_ch1	Channel-specific sleep mode for a Single Channel setup.	Inactive									X								
sleep	Go to sleep-mode.	Inactive								X									
pd	Go to power-down.	Inactive							X										
pd_pin_cfg <1:0>	Configures the PD pin function.	PD pin configured for power-down mode					X	X											0x11
ilvds_lclk <2:0>	LVDS current drive programmability for LCLKP and LCLKN pins.	3.5 mA drive														X	X	X	
ilvds_frame <2:0>	LVDS current drive programmability for FCLKP and FCLKN pins.	3.5 mA drive										X	X	X					
ilvds_dat <2:0>	LVDS current drive programmability for output data pins.	3.5 mA drive					X	X	X										
en_lvds_term	Enables internal termination for LVDS buffers.	Termination disabled		X															0x12
term_lclk <2:0>	Programmable termination for LCLKN and LCLKP buffers.	Termination disabled		1												X	X	X	
term_frame <2:0>	Programmable termination for FCLKN and FCLKP buffers.	Termination disabled		1								X	X	X					
term_dat <2:0>	Programmable termination for output data buffers.	Termination disabled		1			X	X	X										
invert4_ch <4:1>	Channel specific swapping of the analog input signal for a Quad Channel setup.	IPx is positive input													X	X	X	X	0x24
invert2_ch <2:1>	Channel specific swapping of the analog input signal for a Dual Channel setup.	IPx is positive input											X	X					
invert1_ch1	Channel specific swapping of the analog input signal for a Single Channel setup.	IPx is positive input										X							
en_ramp	Enables a repeating full-scale ramp pattern on the outputs.	Inactive										X	0	0					0x25
dual_custom_pat	Enable the mode wherein the output toggles between two defined codes.	Inactive										0	X	0					
single_custom_pat	Enables the mode wherein the output is a constant specified code.	Inactive										0	0	X					
bits_custom1 <15:0>	Bits for the single custom pattern and for the first code of the dual custom pattern. <0> is the LSB.	0x0000	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	0x26



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

Table 5: Register Map

Name	Description	Default	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Hex Address
bits_custom2 <15:0>	Bits for the second code of the dual custom pattern.	0x0000	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	0x27
cgain4_ch1 <3:0>	Programmable coarse gain channel 1 in a Quad Channel setup.	1x gain													X	X	X	X	0x2A
cgain4_ch2 <3:0>	Programmable coarse gain channel 2 in a Quad Channel setup.	1x gain									X	X	X	X					
cgain4_ch3 <3:0>	Programmable coarse gain channel 3 in a Quad Channel setup.	1x gain					X	X	X	X									
cgain4_ch4 <3:0>	Programmable coarse gain channel 4 in a Quad Channel setup.	1x gain	X	X	X	X													
cgain2_ch1 <3:0>	Programmable coarse gain channel 1 in a Dual Channel setup.	1x gain													X	X	X	X	0x2B
cgain2_ch2 <3:0>	Programmable coarse gain channel 2 in a Dual Channel setup.	1x gain									X	X	X	X					
cgain1_ch1 <3:0>	Programmable coarse gain channel 1 in a Single Channel setup.	1x gain					X	X	X	X									
jitter_ctrl <7:0>	Clock jitter adjustment.	160 fsrms									X	X	X	X	X	X	X	X	
precision_mode *	Enable Quad Channel 14 bits precision mode.	Inactive													X				0x31
high_speed_mode *	Enable high speed mode, Single, Dual or Quad channel.	High speed mode – Quad Channel														X	X	X	
clk_divide <1:0> *	Define clock divider factor: 1, 2, 4 or 8	Divide by 1							X	X									
coarse_gain_cfg	Configures the coarse gain setting	x-gain enabled																X	0x33
fine_gain_en	Enable use of fine gain.	Disabled															X		
fgain_branch1 <6:0>	Programmable fine gain for branch 1.	1x / 0dB gain										X	X	X	X	X	X	X	0x34
fgain_branch2 <6:0>	Programmable fine gain for branch 2.	1x / 0dB gain		X	X	X	X	X	X	X									
fgain_branch3 <6:0>	Programmable fine gain for branch 3.	1x / 0dB gain										X	X	X	X	X	X	X	0x35
fgain_branch4 <6:0>	Programmable fine gain for branch 4.	1x / 0dB gain		X	X	X	X	X	X	X									
fgain_branch5 <6:0>	Programmable fine gain for branch 5.	1x / 0dB gain										X	X	X	X	X	X	X	0x36
fgain_branch6 <6:0>	Programmable fine gain for branch 6.	1x / 0dB gain		X	X	X	X	X	X	X									
fgain_branch7 <6:0>	Programmable fine gain for branch 7.	1x / 0dB gain										X	X	X	X	X	X	X	0x37
fgain_branch8 <6:0>	Programmable fine gain for branch 8.	1x / 0dB gain		X	X	X	X	X	X	X									
inp_sel_adc1 <4:0>	Input select for adc 1.	Signal input: IP1/ IN1												X	X	X	X	X	0x3A
inp_sel_adc2 <4:0>	Input select for adc 2.	Signal input: IP2/ IN2				X	X	X	X	X									



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

Table 5: Register Map

Name	Description	Default	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Hex Address
inp_sel_adc3 <4:0>	Input select for adc 3.	Signal input: IP3/ IN3												X	X	X	X	X	0x3B
inp_sel_adc4 <4:0>	Input select for adc 4.	Signal input: IP4/ IN4				X	X	X	X	X									
phase_ddr <1:0>	Controls the phase of the LCLK output relative to data.	90 degrees										X	X						0x42
pat_deskew	Enable deskew pattern mode.	Inactive															0	X	0x45
pat_sync	Enable sync pattern mode.	Inactive															X	0	
btc_mode	Binary two's complement format for ADC output data.	Straight offset binary														X			0x46
msb_first	Serialized ADC output data comes out with MSB first.	LSB first													X				
adc_curr <2:0>	ADC current scaling.	Nominal														X	X	X	0x50
ext_vcm_bc <1:0>	VCM buffer driving strength control.	Nominal											X	X					
lvds_pd_ mode	Controls LVDS power down mode	High z-mode																X	0x52
lvds_output_ mode <2:0> *	Sets the number of LVDS output bits.	12 bit														X	X	X	0x53
low_clk_ freq *	Low clock frequency used.	Inactive													X				
lvds_ advance	Advance LVDS data bits and frame clock by one clock cycle	Inactive											0	X					
lvds_delay	Delay LVDS data bits and frame clock by one clock cycle	Inactive											X	0					
fs_ctrl <5:0>	Fine adjust ADC full scale range	0% change											X	X	X	X	X	X	0x55
startup_ctrl <2:0> *	Controls start-up time.	'000'														X	X	X	0x56

Undefined register addresses must not be written to; incorrect behavior may be the result.

Unused register bits (blank table cells) must be set to '0' when programming the registers.

All registers can be written to while the chip is in power down.

* These registers requires a power down cycle when written to (See Start up Initialization).



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

Register Description

Software Reset

Name	Description	Default	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Hex Address
rst	Self-clearing software reset.	Inactive																X	0x00

Setting the rst register bit to '1', restores the default value of all the internal registers including the rst register bit itself.

Modes of Operation

Name	Description	Default	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Hex Address
precision_mode	Enable Quad Channel 14 bits precision mode.	Inactive													X				0x31
high_speed_mode <2:0>	Enable high speed mode, Single, Dual or Quad channel.	High speed mode – Quad Channel														X	X	X	
clk_divide<1:0>	Define clock divider factor: 1, 2, 4 or 8	Divide by 1							X	X									

The HMCAD1520 has four main operating modes controlled by the register bits precision_mode and high_speed_mode as defined in table 6. Power down mode, as described in section 'Startup Initialization', must be activated after or during a change of operating mode to ensure correct operation. The high speed modes all utilize interleaving to achieve high sampling speed. Quad channel mode interleaves 2 ADC branches, dual channel mode interleaves 4 ADC branches, while single channel mode interleave all 8 ADC branches. In precision mode interleaving is not required and each ADC channel uses one ADC branch only.

Table 6: Modes of Operation

precision_mode	high_speed_mode <2:0>			Mode of operation	Description
0	0	0	1	Single channel 12-bit high speed mode	Single channel by interleaving ADC1 to ADC4
0	0	1	0	Dual channel 12-bit high speed mode	Dual channel where channel 1 is made by interleaving ADC1 and ADC2, channel 2 by interleaving ADC3 and ADC4
0	1	0	0	Quad channel 12-bit high speed mode	Quad channel where channel 1 corresponds to ADC1, channel2 to ADC2, channel3 to ADC3 and channel 4 to ADC4
1	0	0	0	Quad channel 14-bit precision mode	Quad channel where channel 1 corresponds to ADC1, channel2 to ADC2, channel3 to ADC3 and channel 4 to ADC4

Only one of the 4 bits should be activated at the same time.

clk_divide<1:0> allows the user to apply an input clock frequency higher than the sampling rate. The clock divider will divide the input clock frequency by a factor of 1, 2, 4, or 8, defined by the clk_divide<1:0> register. By setting the clk_divide<1:0> value relative to the channel_num<2:0> value, the same input clock frequency can be used for all settings on number of channels. e.g: When increasing the number of channels from 1 to 4, the maximum sampling rate is reduced by a factor of 4. By letting clk_divide<1:0> follow the channel_num<2:0> value, and change it from 1 to 4, the internal clock divider will provide the reduction of the sampling rate without changing the input clock frequency.



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

Table 7: Clock Divider Factor

clk_divide<1:0>	Clock Divider Factor	Sampling rate (FS)
00 (default)	1	Input clock frequency / 1
01	2	Input clock frequency / 2
10	4	Input clock frequency / 4
11	8	Input clock frequency / 8

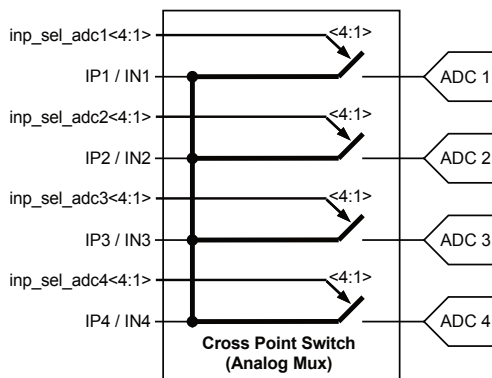
Input Select

Name	Description	Default	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Hex Address
inp_sel_adc1<4:0>	Input select for adc 1.	Signal input: IP1/IN1												X	X	X	X	0	0x3A
inp_sel_adc2<4:0>	Input select for adc 2.	Signal input: IP2/IN2				X	X	X	X	0									
inp_sel_adc3<4:0>	Input select for adc 3.	Signal input: IP3/IN3												X	X	X	X	0	0x3B
inp_sel_adc4<4:0>	Input select for adc 4.	Signal input: IP4/IN4				X	X	X	X	0									

Each ADC is connected to the four input signals via a full flexible cross point switch, set up by inp_sel_adcx. In single channel mode, any one of the four inputs can be selected as valid input to the single ADC channel. In dual channel mode, any two of the four inputs can be selected to each ADC channel. In quad channel mode and precision mode, any input can be assigned to any ADC channel. The switching of inputs can be done during normal operation, and no additional actions are needed. The switching will occur instantaneously at the end of each SPI command.

Table 8: ADC Input Select

inp_sel_adcx<4:0>	Selected input
0001 0	IP1/IN1
0010 0	IP2/IN2
0100 0	IP3/IN3
1000 0	IP4/IN4
other	Do not use


Figure 10: ADC input signals through Cross Point Switch



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

Full-Scale Control

Name	Description	Default	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Hex Address
fs_cntrl <5:0>	Fine adjust ADC full scale range	0% change											X	X	X	X	X	X	0x55

The full-scale voltage range of HMCAD1520 can be adjusted using an internal 6-bit DAC controlled by the fs_cntrl register. Changing the value in the register by one step, adjusts the full-scale range by approximately 0.3%. This leads to a maximum range of $\pm 10\%$ adjustment. Table 9 shows how the register settings correspond to the full-scale range. Note that the values for full-scale range adjustment are approximate. The DAC is, however, guaranteed to be monotonous.

The full-scale control and the programmable gain features differ in two major ways:

1. The full-scale control function is an analog, whereas the programmable gain is a digital function.
2. The programmable gain function has much coarser gain steps and larger range compared to the full-scale control function.

**Table 9: Register Values with
Corresponding Change in Full-Scale Range**

fs_cntrl<5:0>	Full-scale range adjustment
111111	9.7 %
111110	9.4 %
100001	0.3 %
100000	0%
011111	-0.3 %
000001	-9.7%
000000	-10.0%

Current Control

Name	Description	Default	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Hex Address
adc_curr <2:0>	ADC current scaling.	Nominal														X	X	X	0x50
ext_vcm_bc <1:0>	VCM buffer driving strength control	Nominal											X	X					

There are two registers that impact performance and power dissipation.

The *adc_curr* register scales the current consumption in the ADC core. The performance is guaranteed at the nominal setting. Lower power consumption can be achieved by reducing the adc_curr value, see table 10. The impact on performance will depend on the ADC sampling rate.



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

Table 10: ADC Current Control Settings

adc_curr<2:0>	ADC core current
100	-40%
101	-30%
110	-20%
111	-10%
000 (default)	Nominal
001	Do not use
010	Do not use
011	Do not use

The ext_vcm_bc register controls the driving strength in the buffer supplying the voltage on the VCM pin. If this pin is not in use, the buffer can be switched off. If current is drawn from the VCM pin, the driving strength can be increased to keep the voltage on this pin at the correct level.

**Table 11: External Common
Mode Voltage Buffer Driving Strength**

ext_vcm_bc<1:0>	VCM buffer driving strength [μA] Max current sunk/sourced from VCM pin with < 20 mV voltage change.
00	Off (VCM floating)
01 (default)	±20
10	±400
11	±700

Start-up and Clock Jitter Control

Name	Description	Default	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Hex Address
startup_ctrl <2:0>	Controls start-up time	'000'														X	X	X	0x56
jitter_ctrl <7:0>	Clock jitter adjustment	160 fsrms									X	X	X	X	X	X	X	X	0x30

To optimize start up time, a register is provided where the start-up time in clock cycles can be set. Some internal circuitry have start up times that are clock frequency independent. Default counter values are set to accommodate these start up times at the maximum clock frequency (sampling rate). This will lead to increased start up times at low clock frequencies. Setting the value of this register to the nearest higher clock frequency will reduce the count values of the internal counters, to better fit the actual start up time, such that the start up time will be reduced. The start up times from power down and sleep modes are changed by this register setting. If the clock divider is used (set to other than 1), the input clock frequency must be divided by the divider factor to find the correct clock frequency range (see table 7).



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

Table 12: Start-Up Time Control Settings

Precision mode				Quad channel – High speed			
startup_ctrl<2:0>	Clock frequency range [MSPS]	Startup delay [clock cycles]	Startup delay [μs]	startup_ctrl<2:0>	Clock frequency range [MSPS]	Startup delay [clock cycles]	Startup delay [μs]
100	80 - 125	1536	12.3 - 19.2	100	160 - 250	3072	12.3 – 19.2
000	50 - 80	992	12.4 - 19.8	000	100 - 160	1984	12.4 - 19.8
001	32,5 - 50	640	12.8 - 19.7	001	65 - 100	1280	12.8 - 19.7
010	20 - 32,5	420	12.9 - 21	010	40 - 65	840	12.9 - 21
011	15 - 20	260	13 - 17.3	011	30 - 40	520	13 - 17.3
other	Do not use	-	-	other	Do not use	-	-

Dual channel – High speed				Single channel – High speed			
startup_ctrl<2:0>	Clock frequency range [MSPS]	Startup delay [clock cycles]	Startup delay [μs]	startup_ctrl<2:0>	Clock frequency range [MSPS]	Startup delay [clock cycles]	Startup delay [μs]
100	320 - 500	6144	12.3 – 19.2	100	640 - 1000	12288	12.3 – 19.2
000	200 - 320	3968	12.4 - 19.8	000	400 - 640	7936	12.4 - 19.8
001	130 – 200	2560	12.8 - 19.7	001	260 - 400	5120	12.8 - 19.7
010	80 - 130	1680	12.9 - 21	010	160 - 260	3360	12.9 - 21
011	60 – 80	1040	13 - 17.3	011	120 - 160	2080	13 - 17.3
other	Do not use	-	-	other	Do not use	-	-

jitter_ctrl<7:0> allows the user to set a trade-off between power consumption and clock jitter. If all bits in the register is set low, the clock signal is stopped. The clock jitter depends on the number of bits set to '1' in the *jitter_ctrl<7:0>* register. which bits are set high does not affect the result.

Table 13: Clock Jitter Performance

Number of bits to '1' in jitter_ctrl<7:0>	Clock jitter performance Precision mode [fsrms]	Clock jitter performance High speed modes [fsrms]	Module current consumption [mA]
1	130	160	1
2	100	150	2
3	92	136	3
4	85	130	4
5	82	126	5
6	80	124	6
7	77	122	7
8	75	120	8
0	Clock stopped	Clock stopped	



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

LVDS Output Configuration and Control

Name	Description	Default	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Hex Address
lvds_output_mode <2:0>	Sets the number of LVDS output bits.	12 bit														X	X	X	0x53
low_clk_freq	Low clock frequency used.	Inactive													X				
lvds_advance	Advance LVDS data bits and frame clock by one clock cycle	Inactive											0	X					
lvds_delay	Delay LVDS data bits and frame clock by one clock cycle	Inactive											X	0					
phase_ddr<1:0>	Controls the phase of the LCLK output relative to data.	90 degrees										X	X						0x42
btc_mode	Binary two's complement format for ADC output data.	Straight offset binary														X			0x46
msb_first	Serialized ADC output data comes out with MSB first.	LSB first													X				

The HMCAD1520 serial LVDS output has four different modes selected by the register *lvds_output_mode* as defined in table 14. Power down mode, as described in section 'Startup Initialization', must be activated after or during a change in the number of output bits to ensure correct behavior.

Table 14: Number of Bits in LVDS Output

lvds_output_mode <2:0>	Number of Bits	Comment
000	8 bit	8 bit mode, up to 1 GSPS (See HMCAD1511 datasheet)
001	12 bit	Recommended setting for High Speed Modes (Default)
101	14 bit	Recommended setting up to 70 MSPS (Precision mode)
011	16 bit	
100	Dual 8 bit	Recommended setting above 70 MSPS (Precision mode)
Other	Do not use	

12-bit LVDS mode is default for all operational modes. If another LVDS mode is to be used, the *lvds_output_mode* register setting must be changed accordingly.

When 8-bit LVDS mode is used, the LSBs are truncated and the data output will have 8-bit resolution. See HMCAD1511 and HMCAD1510 for detailed description.

When 14 or 16 bit LVDS output mode is selected the output data will be a 13 bit left justified word filled up with '0's on the LSB side. The different high speed modes uses the LVDS outputs as defined by table 15.



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

Table 15: High Speed Modes and Use of LVDS Outputs

High speed modes/ channels	LVDS outputs used
Single channel	D1A, D1B, D2A, D2B, D3A, D3B, D4A, D4B
Dual channel, channel 1	D1A, D1B, D2A, D2B
Dual channel, channel 2	D3A, D3B, D4A, D4B
Quad channel, channel 1	D1A, D1B
Quad channel, channel 2	D2A, D2B
Quad channel, channel 3	D3A, D3B
Quad channel, channel 4	D4A, D4B

For the 14-bit precision mode 14, 16 or dual 8-bit LVDS mode should be used. If the default 12-bit LVDS mode is used, the data output will be truncated to 12 bit. If the 16-bit LVDS mode is used the data output will be a 14-bit left justified word filled up with '00' on the LSB side. If the dual 8-bit output mode is used the 8 most significant bit of the 14 bit data word will be available on the LVDS 'A' output and the remaining 6 bit will be left justified and filled up with '00' on the LVDS 'B' output, see table 16.

Table 16: Precision Mode and Use of LVDS Outputs

Precision mode	LVDS outputs used
Channel 1 - 12, 14, 16-bit output	D1A (D1B will be in power down – high Z)
Channel 1 - Dual 8-bit output	D1A, D1B
Channel 2 - 12, 14, 16-bit output	D2A (D2B will be in power down – high Z)
Channel 2 - Dual 8-bit output	D2A, D2B
Channel 3 - 12, 14, 16-bit output	D3A (D3B will be in power down – high Z)
Channel 3 - Dual 8-bit output	D3A, D3B
Channel 4 - 12, 14, 16-bit output	D4A (D4B will be in power down – high Z)
Channel 4 - Dual 8-bit output	D4A, D4B

Maximum data output bit-rate for the HMCAD1520 is 1 Gb/s. The maximum sampling rate for the different configurations is given by table 17. The sampling rate is set by the frequency of the input clock (FS). The frame-rate, i.e. the frequency of the FCLK signal on the LVDS outputs, depends on the selected mode and the sampling frequency (FS) as defined in table 18.

Table 17: Maximum Sampling Rate vs Number of Output Bits for Different HMCAD1520 Configurations

Number of bits	Single Channel High Speed [MSPS]	Dual Channel High Speed [MSPS]	Quad Channel High Speed [MSPS]	Quad Channel Precision [MSPS]
8	1000	500	250	-
12	660	330	165	82.5
14	560	280	140	70
16	500	250	125	62.5
Dual 8	-	-	-	125



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

Table 18: Output Data Frame Rate

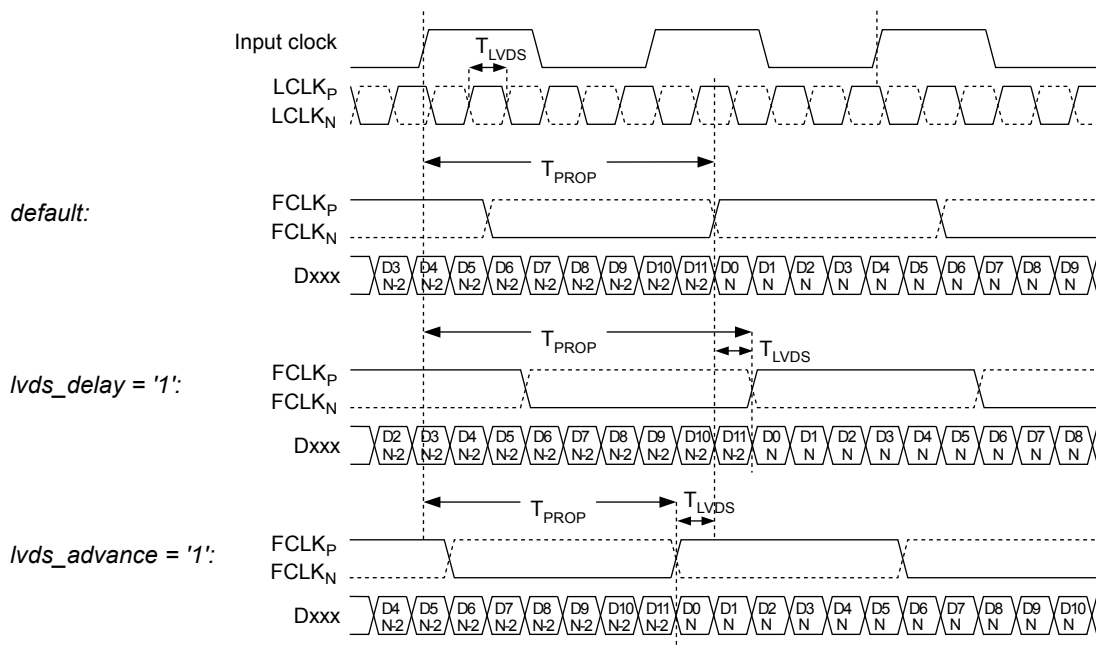
Mode of operation	Frame-rate (FCLK frequency)
High speed, single channel	$F_s / 8$
High speed, dual channel	$F_s / 4$
High speed, quad channel	$F_s / 2$
Precision mode	F_s

If the HMCAD1520 device is used at a low sampling rate the register bit `low_clk_freq` has to be set to '1'. See table 19 for when to use this register bit for the different modes of operation.

Table 19: Use of Register Bit `low_clk_freq`

Mode of operation	Limit when <code>low_clk_freq</code> should be activated
High speed, single channel	$F_s < 240$ MHz
High speed, dual channel	$F_s < 120$ MHz
High speed, quad channel	$F_s < 60$ MHz
Precision mode	$F_s < 30$ MHz

To ease timing in the receiver when using multiple HMCAD1520, the device has the option to adjust the timing of the output data and the frame clock. The propagation delay with respect to the ADC input clock can be moved one LVDS clock cycle forward or backward, by using `lvds_delay` and `lvds_advance`, respectively. See figure 11 for details. Note that LCLK is not affected by `lvds_delay` or `lvds_advance` settings.


Figure 11: LVDS output timing adjustment



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

The LVDS output interface of HMCAD1520 is a DDR interface. The default setting is with the LCLK rising and falling edge transitions in the middle of alternate data windows. The phase for LCLK can be programmed relative to the output frame clock and data bits using `phase_dds<1:0>`. The LCLK phase modes are shown in figure 12. The default timing is identical to setting `phase_dds<1:0>='10'`.

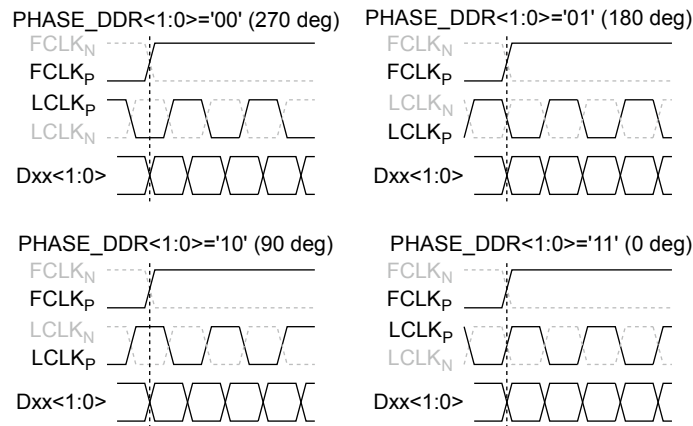


Figure 12: Phase programmability modes for LCLK

The default data output format is offset binary. Two's complement mode can be selected by setting the `btc_mode` bit to '1' which inverts the MSB.

The first bit of the frame (following the rising edge of FCLKP) is the LSB of the ADC output for default settings. Programming the `msb_first` mode results in reverse bit order, and the MSB is output as the first bit following the FCLKP rising edge.

LVDS Drive Strength Programmability

Name	Description	Default	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Hex Address
<code>ilvds_lclk<2:0></code>	LVDS current drive programmability for LCLKP and LCLKN pins.	3.5 mA drive														X	X	X	0x11
<code>ilvds_frame<2:0></code>	LVDS current drive programmability for FCLKP and FCLKN pins.	3.5 mA drive										X	X	X					
<code>ilvds_dat<2:0></code>	LVDS current drive programmability for output data pins.	3.5 mA drive						X	X	X									

The current delivered by the LVDS output drivers can be configured as shown in table 20. The default current is 3.5mA, which is what the LVDS standard specifies.

The LVDS interface offers good robustness at the RSDS (Reduced Swing Differential Signaling), given a careful LVDS wire layout. Using the 1.5mA RSDS will reduce the power consumption significantly compared to default 3.5mA LVDS.

Setting the `ilvds_lclk<2:0>` register controls the current drive strength of the LVDS clock output on the LCLKP and LCLKN pins.

Setting the `ilvds_frame<2:0>` register controls the current drive strength of the frame clock output on the FCLKP and FCLKN pins.

Setting the `ilvds_dat<2:0>` register controls the current drive strength of the data outputs on the D[8:1]P and D[8:1]N pins.



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

Table 20: LVDS Output Drive Strength for LCLK, FCLK and Data

ilvds_*<2:0>	LVDS Drive Strength
000	3.5 mA (default)
001	2.5 mA
101	1.5 mA (RSDS)
011	0.5 mA
100	7.5 mA
101	6.5 mA
110	5.5 mA
111	4.5 mA

LVDS Internal Termination Programmability

Name	Description	Default	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Hex Address
en_lvds_term	Enables internal termination for LVDS buffers.	Termination disabled		X															0x12
term_lclk<2:0>	Programmable termination for LCLKN and LCLKP buffers.	Termination disabled		1												X	X	X	
term_frame<2:0>	Programmable termination for FCLKN and FCLKP buffers.	Termination disabled		1								X	X	X					
term_dat<2:0>	Programmable termination for output data buffers.	Termination disabled		1				X	X	X									

The off-chip load on the LVDS buffers may represent a characteristic impedance that is not perfectly matched with the PCB traces. This may result in reflections back to the LVDS outputs and loss of signal integrity. This effect can be mitigated by enabling an internal termination between the positive and negative outputs of each LVDS buffer. Internal termination mode can be selected by setting the *en_lvds_term* bit to '1'. Once this bit is set, the internal termination values for the bit clock, frame clock, and data buffers can be independently programmed using sets of three bits. Table 21 shows how the internal termination of the LVDS buffers are programmed. The values are typical values and can vary by up to $\pm 20\%$ from device to device and across temperature.

Table 21: LVDS Output Internal Termination for LCLK, FCLK and Data

term_*<2:0>	LVDS Internal Termination
000	Termination disabled
001	260 Ω
010	150 Ω
011	94 Ω
100	125 Ω
101	80 Ω
110	66 Ω
111	55 Ω



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

Power Mode Control

Name	Description	Default	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Hex Address
sleep4_ch<4:1>	Channel-specific sleep mode for a Quad Channel setup.	Inactive													X	X	X	X	0x0F
sleep2_ch<2:1>	Channel-specific sleep mode for a Dual Channel setup.	Inactive											X	X					
sleep1_ch1	Channel-specific sleep mode for a Single Channel setup.	Inactive										X							
sleep	Go to sleep-mode.	Inactive								X									
pd	Go to power-down.	Inactive							X										
pd_pin_cfg<1:0>	Configures the PD pin function.	PD pin configured for power-down mode					X	X											
lvds_pd_mode	Controls LVDS power down mode	High z-mode																X	0x52

The HMCAD1520 device has several modes for power management, from sleep modes with short start up time to full power down with extremely low power dissipation. There are two sleep modes, both with the LVDS clocks (FCLK, LCLK) running, such that the synchronization with the receiver is maintained. The first is a light sleep mode (sleep*_ch) with short start up time, and the second a deep sleep mode (sleep) with the same start up time as full power down.

Setting sleep4_ch<n> = '1' sets channel <n> in a Quad Channel setup in sleep mode, setting sleep2_ch<n> = '1' sets channel <n> in a Dual Channel setup in sleep mode and at last setting sleep1_ch1 = '1' sets the ADC channel in a Single Channel setup in sleep mode. This is a light sleep mode with short start up time.

Setting sleep = '1', puts all channels to sleep, but keeps FCLK and LCLK running to maintain LVDS synchronization. The start up time is the same as for complete power down. Power consumption is significantly lower than for setting all channels to sleep by using the sleep*_Channel register.

Setting pd = '1' completely powers down the chip, including the band-gap reference circuit. Start-up time from this mode is significantly longer than from the sleep*_Channel mode. The synchronization with the LVDS receiver is lost since LCLK and FCLK outputs are put in high-Z mode.

Setting pdn_pin_cfg<1:0> = 'x1' configures the circuit to enter sleep channel mode (all channels off) when the PD pin is set high. This is equal to setting all channels to sleep by using sleep*_ch. The channels can not be powered down separately using the PD pin. Setting pdn_pin_cfg<1:0> = '10' configures the circuit to enter (deep) sleep mode when the PD pin is set high (equal to setting sleep='1'). When pdn_pin_cfg <1:0> = '00', which is the default, the circuit enters the power down mode when the PD pin is set high.

The lvds_pd_mode register configures whether the LVDS data output drivers are powered down or kept alive in sleep and sleep channel modes. LCLK and FCLK drivers are not affected by this register, and are always on in sleep and sleep channel modes. If lvds_pd_mode is set low (default), the LVDS output is put in high Z mode, and the driver is completely powered down. If lvds_pd_mode is set high, the LVDS output is set to constant 0, and the driver is still on during sleep and sleep channel modes.



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

Programmable Gain

Name	Description	Default	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Hex Address
coarse_gain_cfg	Configures the coarse gain setting	x-gain enabled																X	0x33
fine_gain_en	Enable use of fine gain.	Disabled															X		
cgain4_ch1<3:0>	Programmable coarse gain channel 1 in a Quad Channel setup.	1x gain													X	X	X	X	0x2A
cgain4_ch2<3:0>	Programmable coarse gain channel 2 in a Quad Channel setup.	1x gain									X	X	X	X					
cgain4_ch3<3:0>	Programmable coarse gain channel 3 in a Quad Channel setup.	1x gain					X	X	X	X									
cgain4_ch4<3:0>	Programmable coarse gain channel 4 in a Quad Channel setup.	1x gain	X	X	X	X													
cgain2_ch1<3:0>	Programmable coarse gain channel 1 in a Dual Channel setup.	1x gain													X	X	X	X	0x2B
cgain2_ch2<3:0>	Programmable coarse gain channel 2 in a Dual Channel setup.	1x gain									X	X	X	X					
cgain1_ch1<3:0>	Programmable coarse gain channel 1 in a Single Channel setup.	1x gain					X	X	X	X									
fgain_branch1<6:0>	Programmable fine gain for branch1.	1x / 0dB gain										X	X	X	X	X	X	X	0x34
fgain_branch2<6:0>	Programmable fine gain for branch 2.	1x / 0dB gain		X	X	X	X	X	X	X									
fgain_branch3<6:0>	Programmable fine gain for branch 3.	1x / 0dB gain										X	X	X	X	X	X	X	0x35
fgain_branch4<6:0>	Programmable fine gain for branch 4.	1x / 0dB gain		X	X	X	X	X	X	X									
fgain_branch5<6:0>	Programmable fine gain for branch 5.	1x / 0dB gain										X	X	X	X	X	X	X	0x36
fgain_branch6<6:0>	Programmable fine gain for branch 6.	1x / 0dB gain		X	X	X	X	X	X	X									
fgain_branch7<6:0>	Programmable fine gain for branch 7.	1x / 0dB gain										X	X	X	X	X	X	X	0x37
fgain_branch8<6:0>	Programmable fine gain for branch 8.	1x / 0dB gain		X	X	X	X	X	X	X									

The device includes a digital programmable gain in addition to the Full-scale control. The programmable gain of each channel can be individually set using a four bit code, indicated as cgain* $\langle$ 3:0 $\rangle$. The gain is configured by the register cgain_cfg, when cgain_cfg equals '0' a gain in dB steps is enabled as defined in table 22 otherwise if cgain_cfg equals '1' the gain is defined by table 23.



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

Table 22: Gain Setting – dB step

cgain_cfg	cgain* $\langle$ 3:0 $\rangle$	Implemented gain [dB]
0	0000	0
0	0001	1
0	0010	2
0	0011	3
0	0100	4
0	0101	5
0	0110	6
0	0111	7
0	1000	8
0	1001	9
0	1010	10
0	1011	11
0	1100	12
0	1101	Not used
0	1110	Not used
0	1111	Not used

Table 23: Gain Setting – x step

cgain_cfg	cgain* $\langle$ 3:0 $\rangle$	Implemented gain factor [x]
1	0000	1
1	0001	1.25
1	0010	2
1	0011	2.5
1	0100	4
1	0101	5
1	0110	8
1	0111	10
1	1000	12.5
1	1001	16
1	1010	20
1	1011	25
1	1100	32
1	1101	50
1	1110	Not used
1	1111	Not used

There is a digital fine gain implemented for each ADC branch to adjust the fine gain errors between the branches. The gain is controlled by *fgain_branch** as defined in table 24. For the high speed interleaved modes, there will be no missing codes when using digital fine gain, due to higher resolution internally (1 bit).

To enable the fine gain function the register bit *fine_gain_en* has to be activated, set to ‘1’.



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

Table 24: Fine Gain Setting

fgain_branchx<6:0>							Arithmetic Function	Implemented Gain (x)	Gain (dB)
0	1	1	1	1	1	1	$OUT = (1 + 2^{-8} + 2^{-9} + 2^{-10} + 2^{-11} + 2^{-12} + 2^{-13}) * IN$	1.0077	0.0665
0	1	1	1	1	1	0	$OUT = (1 + 2^{-8} + 2^{-9} + 2^{-10} + 2^{-11} + 2^{-12}) * IN$	1.0076	0.0655
0	1	1	1	1	0	1	$OUT = (1 + 2^{-8} + 2^{-9} + 2^{-10} + 2^{-11} + 2^{-13}) * IN$	1.0074	0.0644
0	1	1	1	1	0	0	$OUT = (1 + 2^{-8} + 2^{-9} + 2^{-10} + 2^{-11}) * IN$	1.0073	0.0634
0	0	0	0	0	1	1	$OUT = (1 + 2^{-12} + 2^{-13}) * IN$	1.0004	0.0031
0	0	0	0	0	1	0	$OUT = (1 + 2^{-12}) * IN$	1.0002	0.0021
0	0	0	0	0	0	1	$OUT = (1 + 2^{-13}) * IN$	1.0001	0.0010
0	0	0	0	0	0	0	$OUT = IN$	1.0000	0.0000
1	1	1	1	1	1	1	$OUT = IN$	1.0000	0.0000
1	1	1	1	1	1	0	$OUT = (1 - 2^{-13}) * IN$	0.9999	-0.0011
1	1	1	1	1	0	1	$OUT = (1 - 2^{-12}) * IN$	0.9998	-0.0021
1	1	1	1	1	0	0	$OUT = (1 - 2^{-12} - 2^{-13}) * IN$	0.9996	-0.0032
1	0	0	0	0	1	1	$OUT = (1 - 2^{-8} - 2^{-9} - 2^{-10} - 2^{-11}) * IN$	0.9927	-0.0639
1	0	0	0	0	1	0	$OUT = (1 - 2^{-8} - 2^{-9} - 2^{-10} - 2^{-11} - 2^{-13}) * IN$	0.9926	-0.0649
1	0	0	0	0	0	1	$OUT = (1 - 2^{-8} - 2^{-9} - 2^{-10} - 2^{-11} - 2^{-12}) * IN$	0.9924	-0.0660
1	0	0	0	0	0	0	$OUT = (1 - 2^{-8} - 2^{-9} - 2^{-10} - 2^{-11} - 2^{-12} - 2^{-13}) * IN$	0.9923	-0.0670

Analog Input Invert

Name	Description	Default	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Hex Address
invert4_ch<4:1>	Channel specific swapping of the analog input signal for a Quad Channel setup.	IPx is positive input													X	X	X	X	0x24
invert2_ch<2:1>	Channel specific swapping of the analog input signal for a Dual Channel setup.	IPx is positive input											X	X					
invert1_ch1	Channel specific swapping of the analog input signal for a Single Channel setup.	IPx is positive input										X							

The IPx pin represents the positive analog input pin, and INx represents the negative (complementary) input. Setting the bits marked *invertx_ch<n:1>* (individual control for each channel) causes the inputs to be swapped. INx would then represent the positive input, and IPx the negative input.



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

LVDS Test Patterns

Name	Description	Default	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	Hex Address
en_ramp	Enables a repeating full-scale ramp pattern on the outputs.	Inactive										X	0	0					0x25
dual_custom_pat	Enable the mode wherein the output toggles between two defined codes.	Inactive										0	X	0					
single_custom_pat	Enables the mode wherein the output is a constant specified code.	Inactive										0	0	X					
bits_custom1<15:0>	Bits for the single custom pattern and for the first code of the dual custom pattern. <0> is the LSB.	0x0000	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	0x26
bits_custom2<15:0>	Bits for the second code of the dual custom pattern.	0x0000	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	0x27
pat_deskew	Enable deskew pattern mode.	Inactive															0	X	0x45
pat_sync	Enable sync pattern mode.	Inactive															X	0	

To ease the LVDS synchronization setup of HMCAD1520, several test patterns can be set up on the outputs. Normal ADC data are replaced by the test pattern in these modes. Setting en_ramp to '1' sets up a repeating full-scale ramp pattern on all data outputs. The ramp starts at code zero and is increased 1LSB every clock cycle. It returns to zero code and starts the ramp again after reaching the full-scale code.

A constant value can be set up on the outputs by setting single_custom_pat to '1', and programming the desired value in bits_custom1<15:0>. In this mode, bits_custom1<15:0> replaces the ADC data at the output, and is controlled by LSB-first and MSB-first modes in the same way as normal ADC data are.

The device may also be set up to alternate between two codes by programming dual_custom_pat to '1'. The two codes are the contents of bits_custom1<15:0> and bits_custom2<15:0>.

Since bit_custom*<15:0> is a 16 bit word there will be a truncation at the LSB side when using less than 16 bits in the LVDS output word. If 12-bit output is selected bit <15:4> will be used, if 14-bit output is used bit <15:2> will be used and if dual 8-bit is selected bit<15:8> will be put on the LVDS 'A' output and bit <7:0> will be put on the LVDS 'B' output.

Two preset patterns can also be selected:

1. Deskew pattern: Set using pat_deskew, this mode replaces the ADC output with a pattern consisting of alternating zeros and ones - MSB will be a zero. For a 12-bit output the pattern will be: '010101010101'
2. Sync pattern: Set using pat_sync, the normal ADC word is in this mode replaced by a fixed synchronization pattern where the output word is split in two and the upper part of the word is ones and the lower part is zeros. For a 12-bit output the pattern will be: '111111000000'.

Note: Only one of the above patterns should be selected at the same time.



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

Theory of Operation

HMCAD1520 is a Multi-Mode high-speed, CMOS ADC, consisting of 8 ADC branches, configured in different channel modes, using interleaving to achieve high speed sampling. For all practical purposes, the device can be considered to contain 4 ADCs. Fine gain is adjusted for each of the eight branches separately.

HMCAD1520 utilizes a LVDS output, described in 'Register Description, LVDS Output Configuration and Control'. The clocks needed (FCLK, LCLK) for the LVDS interface are generated by an internal PLL.

The HMCAD1520 operate from one clock input, which can be differential or single ended. The sampling clocks for each of the four channels are generated from the clock input using a carefully matched clock buffer tree. Internal clock dividers are utilized to control the clock for each ADC during interleaving. The clock tree is controlled by the Mode of operations.

HMCAD1520 uses internally generated references. The differential reference value is 1V. This results in a differential input of -1V to correspond to the zero code of the ADC, and a differential input of +1V to correspond to the maximum code.

The ADC employs a Pipeline converter architecture. each Pipeline Stage feeds its output data into the digital error correction logic, ensuring excellent differential linearity and no missing codes.

HMCAD1520 operates from two sets of supplies and grounds. The analog supply and ground set is identified as AVDD and AVSS, while the digital set is identified by DVDD and DVSS.

Interleaving Effects and Sampling Order

Interleaving ADCs will generate interleaving artifacts caused by gain, offset and timing mismatch between the ADC branches. The design of HMCAD1520 has been optimized to minimize these effects. It is not possible, though, to eliminate mismatch, such that additional compensation may be needed. The internal digital fine gain control may be used to compensate for gain errors between the ADC branches. Due to the optimization of HMCAD1520 there is not a one-to-one correspondence between the sampling order, LVDS output order and the branch number. Tables 25, 26 and 27 give an overview of the corresponding branches, LVDS outputs and sampling order for the different high speed modes.

Table 25: Quad Channel Mode

Channel #	Sampling order	LVDS output	Fine gain branch
1	1	D1A	1
	2	D1B	2
2	1	D2A	3
	2	D2B	4
3	1	D3A	5
	2	D3B	6
4	1	D4A	7
	2	D4B	8

Table 26: Dual Channel Mode

Channel #	Sampling order	LVDS output	Fine gain branch
1	1	D1A	1
	2	D1B	3
	3	D2A	2
	4	D2B	4
2	1	D3A	5
	2	D3B	7
	3	D4A	6
	4	D4B	8

Table 27: Single Channel Mode

Channel #	Sampling order	LVDS output	Fine gain branch
1	1	D1A	1
	2	D1B	6
	3	D2A	2
	4	D2B	5
	5	D3A	8
	6	D3B	3
	7	D4A	7
	8	D4B	4

Precision Mode

In precision mode the resolution of each ADC channel is increased from 12 bits to 14 bits. In order to get the additional performance, the LVDS outputs have to be set up in 14, 16 or dual 8-bit configuration.

When digital fine gain (registers 34-37hex) is used in precision mode, the mapping between ADC channel and ADC branch in table 28 should be used.



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

Table 28: Overview of Fine Gain Usage in Precision Mode

Channel #	LVDS output	Fine gain branch
1	D1A, (D1B)	1
2	D2A, (D2B)	3
3	D3A, (D3B)	5
4	D4A, (D4B)	7

Analog Input

The analog input to HMCAD1520 is a switched capacitor track-and-hold amplifier optimized for differential operation. Operation at common mode voltages at mid supply is recommended even if performance will be good for the ranges specified. The VCM pin provides a voltage suitable as common mode voltage reference. The internal buffer for the VCM voltage can be switched off, and driving capabilities can be changed programming the ext_vcm_bc<1:0> register.

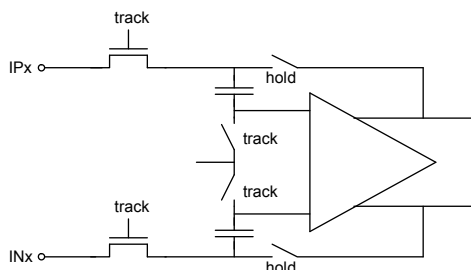


Figure 13: Input configuration

Figure 13 shows a simplified drawing of the input network. The signal source must have sufficiently low output impedance to charge the sampling capacitors within one clock cycle. A small external resistor (e.g. 22 ohm) in series with each input is recommended as it helps reducing transient currents and dampens ringing behavior. A small differential shunt capacitor at the chip side of the resistors may be used to provide dynamic charging currents and may improve performance. The resistors form a low pass filter with the capacitor, and values must therefore be determined by requirements for the application.

DC-Coupling

Figure 14 shows a recommended configuration for DC-coupling. Note that the common mode input voltage must be controlled according to specified values. Preferably, the CM_EXT output should be used as reference to set the common mode voltage.

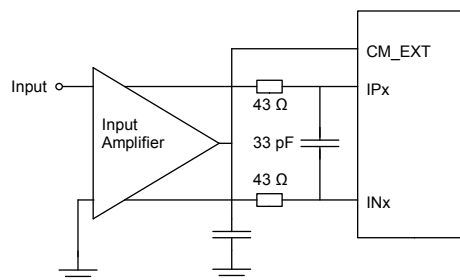


Figure 14: DC coupled input

The input amplifier could be inside a companion chip or it could be a dedicated amplifier. Several suitable single ended to differential driver amplifiers exist in the market. The system designer should make sure the specifications of the selected amplifier is adequate for the total system, and that driving capabilities comply with HMCAD1520 input specifications.

Detailed configuration and usage instructions must be found in the documentation of the selected driver, and the values given in figure 14 must be varied according to the recommendations for the driver.

AC-Coupling

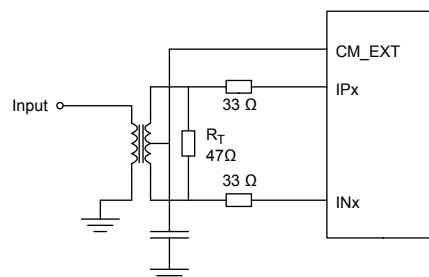


Figure 15: Transformer coupled input

A signal transformer or series capacitors can be used to make an AC-coupled input network. Figure 15 shows a recommended configuration using a transformer. Make sure that a transformer with sufficient linearity is selected, and that the bandwidth of the transformer is appropriate. The bandwidth should exceed the sampling rate of the ADC with at least a factor of 10. It is also important to minimize phase mismatch between the differential ADC inputs for good HD2 performance. This type of transformer coupled input is the preferred configuration for high frequency signals as most differential amplifiers do not have adequate performance at high frequencies.



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

Magnetic coupling between the transformers and PCB traces may impact channel crosstalk, and must hence be taken into account during PCB layout.

If the input signal is traveling a long physical distance from the signal source to the transformer (for example a long cable), kick-backs from the ADC will also travel along this distance. If these kick-backs are not terminated properly at the source side, they are reflected and will add to the input signal at the ADC input. This could reduce the ADC performance. To avoid this effect, the source must effectively terminate the ADC kick-backs, or the traveling distance should be very short.

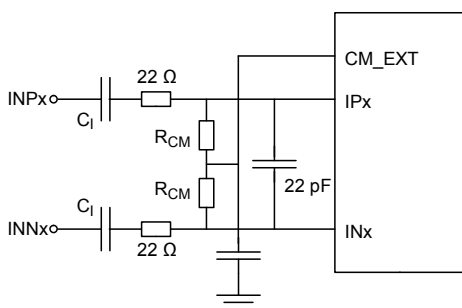


Figure 16: AC coupled input

Figure 16 shows AC-coupling using capacitors. Resistors from the CM_EXT output, RCM, should be used to bias the differential input signals to the correct voltage. The series capacitor, Ci, form the high-pass pole with these resistors, and the values must therefore be determined based on the requirement to the high-pass cut-off frequency.

Note that Start Up Time from Sleep Mode and Power Down Mode will be affected by this filter as the time required to charge the series capacitors is dependent on the filter cut-off frequency.

Clock Input and Jitter Considerations

Typically high-speed ADCs use both clock edges to generate internal timing signals. In HMCAD1520 only the rising edge of the clock is used. Hence, input clock duty cycles between 30% and 70% are acceptable.

The input clock can be supplied in a variety of formats. The clock pins are AC-coupled internally, hence a wide common mode voltage range is accepted. Differential clock sources such as LVDS, LVPECL or differential sine wave can be utilized. LVDS/LVPECL clock signals must be appropriately terminated as close as possible

to the ADC clock input pins. For CMOS inputs, the CLKN pin should be connected to ground, and the CMOS clock signal should be connected to CLKP. CMOS inputs are not recommended above 200MHz. For differential sine wave clock input the amplitude must be at least +/- 0.8 Vpp. No additional configuration is needed to set up the clock source format.

The quality of the input clock is extremely important for high-speed, high-resolution ADCs. The contribution to SNR from clock jitter with a full scale signal at a given frequency is shown in equation 1.

$$SNR_{jitter} = 20 \cdot \log(2 \cdot \pi \cdot f_{IN} \cdot \epsilon_t) \quad (1)$$

where f_{IN} is the signal frequency, and ϵ_t is the total rms jitter measured in seconds. The rms jitter is the total of all jitter sources including the clock generation circuitry, clock distribution and internal ADC circuitry.

For applications where jitter may limit the obtainable performance, it is of utmost importance to limit the clock jitter. This can be obtained by using precise and stable clock references (e.g. crystal oscillators with good jitter specifications) and make sure the clock distribution is well controlled. It might be advantageous to use analog power and ground planes to ensure low noise on the supplies to all circuitry in the clock distribution. It is of utmost importance to avoid crosstalk between the ADC output bits and the clock and between the analog input signal and the clock since such crosstalk often results in harmonic distortion.

The jitter performance is improved with reduced rise and fall times of the input clock. Hence, optimum jitter performance is obtained with LVDS or LVPECL clock with fast edges. CMOS and sine wave clock inputs will result in slightly degraded jitter performance.

If the clock is generated by other circuitry, it should be re-timed with a low jitter master clock as the last operation before it is applied to the ADC clock input.

LVDS output frequencies

The relationship between LVDS bitrate and sampling frequency is:

$$LVDS_{bitrate} = FS/N_b \cdot N_{lvds}$$

Where:

FS is the sampling frequency.

N_{lvds} is number of output bits on the LVDS interface.

N_b is given by:



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

Single channel mode: 8

Dual channel mode: 4

Quad channel mode: 2

If the input clock divider is used FS is given by:

$$FS = F_{clk} / \text{clock divide factor}$$

The LCLK frequency is given by:

$$F_{lclk} = LVDS\text{Bitrate}/2$$

Application Usage Example

This section gives an overview on how HMCAD1520 can be used in an application utilizing all active modes with a single clock source. The example assumes that a low jitter 500MHz clock source is applied. A differential clock should be used, and can be generated from a single ended low jitter crystal oscillator, using a transformer or balun in conjunction with ac-coupling to convert from single ended to differential signal.

Since the resolution is 12 bits in the high speed modes and 14 bits in precision mode, it will be an advantage to set the LVDS outputs to 14 or 16 bits to avoid changing the LVDS interface when going from one of the high speed modes to precision mode or vice versa. The extra bits added in the LSB position of the transferred word can simply be removed in the receiver. In this example 14 bit LVDS is chosen.

Start-up Initialization

The start-up sequence will be as follows:

- Apply power
- Apply reset (RESETN low, then high, or SPI command 0x00 0x0001)
- Set power down (PD pin high or SPI command 0x0F 0x0200)
- Set 14bit LVDS output mode (SPI command 0x53 0x0002)
- Set LVDS bit clock phase (phase_dds, register 0x42) if other than default must be used (depends on the receiver).
- Select operating mode, for instance dual channel high speed mode, and clock divider factor (SPI command 0x31 0x0102).
- Set active mode (PD pin low or SPI command 0x0F 0x0000)
- Select analog inputs, for instance input 1 on channel 1 and input 3 on channel 2 (SPI commands 0x3A 0202 and 0x3B 0808)

Change Mode

When changing operational mode, power down must be activated due to internal synchronization routines. A typical mode change will then be like this:

- Set power down (PD pin high or SPI command 0x0F 0x0200)
- Change mode to for example Single channel mode (SPI command 0x31 0x0001)
- Set active mode (PD pin low or SPI command 0x0F 0x0000)
- Select analog inputs, for instance Input 1 (SPI commands 0x3A 0202 and 0x3B 0202)

Table 29 gives an overview of the operational modes in this example and the SPI commands to apply for each mode.

Table 29: Overview of Operating Modes and Setup Conditions

Operating mode	Sampling speed [MSPS]	Clock divider factor	SPI command for mode selection and clock divider
Single channel	500	1	0x31 0x0001
Dual channel	250	2	0x31 0x0102
Quad channel	125	4	0x31 0x0204
Quad channel Precision	62.5	8	0x31 0x0308

Select Analog Input

When an operational mode is selected, the analog inputs can be changed 'on-the-fly'. To change analog input one merely have to apply the dedicated SPI commands. The change will occur instantaneously at the end of each SPI command.

Table 30: Example of Some Analog Input Selections

Operating mode	Signal input selection	SPI commands
Single channel	IP4/IN4	0x3A 1010, 0x3B 1010
Dual channel	Ch1: IP2/IN2 Ch2: IP3/IN3	0x3A 0404, 0x3B 0808
Quad channel	Ch1: IP4/IN4 Ch2: IP3/IN3 Ch3: IP2/IN2 Ch4: IP1/IN1	0x3A 1008, 0x3B 0402
Quad channel Precision	Ch1: IP1/IN1 Ch2: IP2/IN2 Ch3: IP3/IN3 Ch4: IP4/IN4	0x3A 0204, 0x3B 0810



HIGH SPEED MULTI-MODE 8/12/14-BIT 1000/640/105 MSPS A/D CONVERTER

Outline Drawing

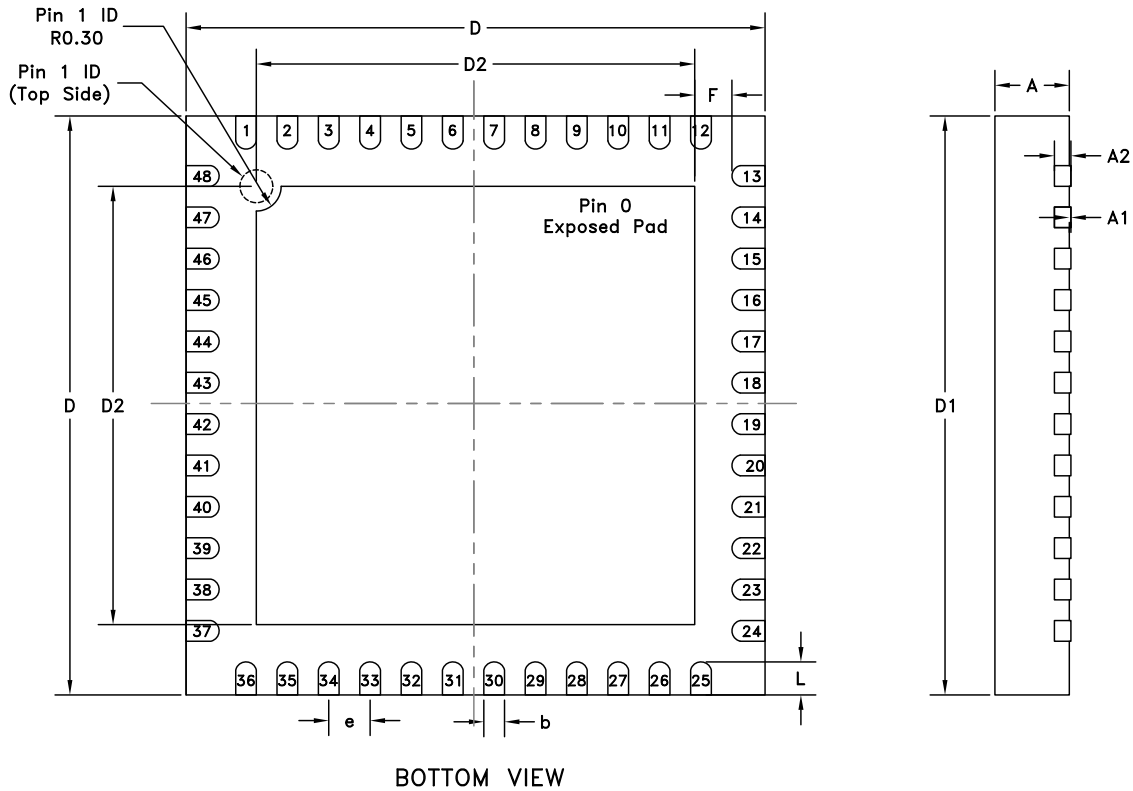


Table 28: 7x7 mm QFN 48 Pin (LP7) Dimensions

Symbol	Millimeter			Inch		
	Min	Typ	Max	Min	Typ	Max
A	0.8	0.9	1	0.031	0.035	0.039
A1	0	0.02	0.05	0	0.0008	0.002
A2		0.2			0.008	
b	0.18	0.25	0.3	0.007	0.01	0.012
D	7.00 bsc			0.276 bsc		
D2	5.15	5.3	5.4	0.203	0.209	0.213
L	0.3	0.4	0.5	0.012	0.016	0.02
e	0.50 bsc			0.020 bsc		
F	0.2			0.008		

Package Information

Part Number	Package Body Material	Lead Finish	MSL [1]	Package Marking [2]
HMCAD1520	RoHS-compliant Low Stress Injection Molded Plastic	100% matte Sn	Level 2A	HAD1520 XXXX

[1] MSL, Peak Temp: The moisture sensitivity level rating classified according to the JEDEC industry standard and to peak solder temperature.

[2] Proprietary marking XXXX, 4-Digit lot number XXXX